DrugArgDet a novel dataset for argument detection in drug reviews

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Abstract

REVIEW - adapt narrative and rephrase The ubiquity of online product reviews has led to a surge in users both posting and reading reviews and comments, seeking valuable insights and opinions from others. Among these, drug reviews stand out as particularly noteworthy, as they often contain crucial factual information gleaned from individual experiences. To effectively mine this information beyond sentiment-based opinion clustering, an argument-mining approach is essential. In this paper, we present the construction of a novel dataset specifically tailored for argumentation in the pharmaceutical domain. We also conduct classification experiments on this dataset, leveraging both traditional machine learning and state-of-the-art transformer models to identify and classify arguments within the drug review context. Our experiments yield promising results, achieving an F1 score of 0.82 in binary classification (argumentative vs. non-argumentative) and 0.76 for three-class classification (argumentative-pro, argumentative-against, and non-argumentative), demonstrating the potential of our approach in effectively mining argumentative structures in the pharmaceutical domain. This work contributes to the growing body of research in the NLP field by providing a unique dataset and a comprehensive analysis of classification techniques for argument mining in drug reviews. Our findings have the potential to unlock valuable insights from user-generated content and facilitate more informed decision-making for consumers, healthcare providers, and pharmaceutical companies alike.

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1. Introduction

The amount of user-generated content on the Internet has increased exponentially during the last decades. Part of this content consists of opinions and reviews about all kinds of products and experiences such as restaurants, travel destinations, films, books, food, drugs, etc. This content provides a rich source of information for other users to make better-informed decisions about their activity, counterbalancing the information provided by sellers and marketing campaigns, e.g. which restaurant to visit or which laptop not to buy. Additionally, it provides sellers and manufacturers an excellent way to obtain feedback, understand people's interaction with their products, and improve their offerings. Drugs and pharmaceutical products are no exception to this trend. Nowadays, we can easily find several specialized websites and fora where users can share their experiences with such products (e.g., Drugs.com, Druglib.com, WebMD.com, RxList, etc.).

The case of drugs is particularly interesting for information extraction due to several reasons. Firstly, drugs are directly applied to the user and could potentially impact their health, even posing risks of death. Secondly, the drug's effectiveness is highly dependent on the consumer's individual circumstances, such as other medical conditions, additional drugs, environmental and genetic factors, etc. Thirdly, healthcare practitioners should prescribe drugs in most cases. Additionally, drugs need to be approved by a committee of experts after passing various scientific studies on different groups of people, among other factors. However, due to the complexity of drug interaction with the human organism, the large variety of human conditions and circumstances, and the lack of control over the actual intake procedure, the results and effects of using these products are sometimes difficult to predict. Consequently, among these drug reviews, we find a vast amount of comments in both directions: positive experiences, where the drug intake produced the desired (or expected) effect; and negative experiences, when something did not go in the direction that the user expected.

Reviews about products usually blend three types of information together: facts, opinions and experiences [1]. While opinions are subjective evaluations or judgments based on an individual's perception, facts are objective and verifiable data that do not depend on each person's perspective. On the other hand, experiences are knowledge extracted from practical contact with and observation of facts or events. When it comes to health products, such as drugs, all three sources of information can be of interest to potential users. However, making informed decisions about health should be based on reliable data and evidence. While a single review of a product may contain subjective information, analyzing a sufficient sample of reviews by multiple users on a specific topic can reveal patterns and extract valuable information. Moreover, in order to extract reliable information from a set of reviews, argumentative pieces of text offer the best quality and most opportunities to uncover the underlying information along with the reasons that support it.

Previous studies have demonstrated the feasibility of analyzing the sentiment of drug reviews to obtain valuable insights, such as sentiment-based opinion groups, drug scoring or ranking, and even predicting the average star ratings assigned by users to a particular drug [2, 3, 4, 5, 6]. However, extracting factual information requires going beyond sentiment analysis and distinguishing opinionated affirmations from evidence-based reasoning or inferred causality related to the experience of taking the drug. Despite personal subjectivity, this reasoning process and inferred causality are essential to understanding the phenomena underlying that personal experience and enable us to connect different user experiences based on common patterns.

Product reviews, especially drug reviews, typically combine two types of discourse: argumentative and non-argumentative. Argumentative elements, regardless of whether they are well-articulated or not, attempt to establish causation or correlation between facts. Non-argumentative texts, on the other hand, are commonly used to offer advice, warnings, and to state beliefs or opinions, without discussing the reasons that support them [7]. Using a suitable argumentation schema, we could identify arguments referring to that drug and its intended use, and we would be able to better understand the nature of that experience, as well as connect that experience to other users' experiences in the same conditions. This qualitative analysis provides improved visibility and explainability to the actual use of these drugs and their effects directly from the users. This direct feedback from the consumers is a rich source of information, as explained in [8]. However, up to date this analysis is usually based on healthcare practitioners' data about consultations or surveys performed by different pharmaceutical companies and health institutions. The few works that have exploited the information in consumers' reviews have focused on detecting the presence of unknown side effects [9].

The first contribution of this work is to propose the use of argument mining techniques to extract relevant information from patient reviews of drugs. This information can be used by potential users to make better-informed decisions about drug consumption and by pharmaceutical companies to understand what is being said about their products, as well as to identify new side effects. Since reviews often contain subjective opinions that may not be factually based, argument extraction techniques can help identify the evidence supporting different claims and facilitate the process of verifying their veracity and credibility.

We have identified several datasets for sentiment analysis tasks in the drug review domain, but none of them are suitable for argument mining. Although there are datasets focusing on argumentation in other domains, the definition of argumentation varies widely and may not be directly applicable to the drug review domain. Therefore, our second contribution is the development of a new dataset specifically for the drug review domain, annotated for argumentation. This dataset can serve as a foundation for tasks involving argument and information mining to extract valuable insights and facilitate further studies, such as the analysis of controversy, opinion dynamics, and the evolution of factual information.

Thirdly, we present a system to automatically identify and classify arguments in user-generated drug opinions. Our approach combines traditional machine learning methods with the state-of-the-art transformer-based models, and is tested on the newly generated dataset of drug reviews. Our approach exploits different types of linguistic, semantic and domain-specific features and demonstrates that it may contribute to the development of more effective pharmacovigilance and drug safety systems, as well as to assist potential user to make better informed decisions.

The rest of the paper is organized as the following: in section 4 we present previous works on related datasets, alongside the details of the DrugArgDet dataset and its development. In section5, we present the different systems used to approach the classification problem. Finally, we present the results and discussion in section6 and our conclusions in section 7.

2. Problem Definition

In the context of drug reviews, drug users describe in online fora their experiences in the use case of specific drugs for a particular condition. Producing and sharing this content online visibilize real experiences with those drugs, providing first-hand impressions to others regarding their experienced effects and side-effects. These data sources have obvious advantages, since they are a first-hand descriptions of the use of the drugs and people can inform themselves about a prescribed medication. Such insightw are difficult to gather through other means, and health institutions and industries rely on surveys and practitioners reporting on results, often when things go wrong. On the other hand, consuming information about personal experiences can undermine trust in medical criteria to prescribe medications, as we have witnessed during the COVID-19 pandemic, undermining the scientific authority of healthcare institutions. This scenario is the so-called *epistemic relativism*: personal and anecdotal experiences are equally considered to scientific evidence in order to take risky – often deadly – decisions. Given these advantages and disadvantages, it seems advisable to analyze these data sources to extract meaningful insights that can throw some light on the reality of those experiences. These texts are focused, as previously described, in specific applications of drug to a medical condition. The texts consist of aspects that worked and didn't work of the drug in that particular use-case. All these aspects are in nature subjective but the degree of subjectivity can vary extensively. As we are interested on extracting elements that are *as close to the evidence as possible*, we need to make a distinction between highly opinionated, sentiment-based, taste-based or irrational aspects from observations and *evidence-based* conclusions derived by the user after applying the drug in the use-case. Not all of these aspects are actually relevant, since they can also be extremely opinionated and express personal preferences and not attempts of *objective* observations or evidence that we could interpret as arguments pro- and against-. There is a need then to distinguish text sections where the user is writing in argumentative manner, specifically about the application of the drug for the condition and nothing else, from those sections that are not relevant for the topic. Once these argumentative pieces are known, we also need to know if they are supporting or attacking such use case of the drug. Having for each review a list of argumentative aspects, both supporting and attacking, and ignoring the *irrelevant* text sections (such as anecdotes, personal life details, personal preferences, opinionated content, etc.) would definitely provide a *near-objective* source of information and better understanding of the applications of such drugs. Hence, the need for the set of tools provided by frameworks in the field of *argument mining, retrieval and detection...*

In contrast to other online user produced reviews, e.g. hotel reviews, product reviews in *Amazon*, etc., drug reviews present At the same time, argumentation could

3. Previous Work

3.1. Argument Mining

Argument mining, as described in [10], argument mining is typically divided into three main sub-tasks in the literature: (i) differentiating argumentative from non-argumentative text units, (ii) classifying argument components or argumentation schemes, and (iii) identifying argumentation structures.

The first subtask of argument mining, separating argumentative from non-argumentative texts, has been addressed in previous works such as Moens and Steedman [11] and Florou et al. [12]. Typically, this problem is tackled as a binary classification task in which text units are classified as either "argumentative" or "non-argumentative". Moens and Steedman used the *Araucaria corpus* [13], which was annotated based on domain-independent argumentation theory [14], to classify text units (sentences, in this case) using various features (including lexical, syntactic, semantic and discourse properties of the texts), achieving more than 73% of accuracy in the identification of argumentative sentences. Similarly, Florou et al. [12] classified text segments (consisting of one or more consecutive sentences with the same polarity) as containing argumentation or not, based on features that focused on tense and mood as well as the presence of discourse markers (such as "despite", "however", etc). They obtained around 73% of F-measure in the binary classification task. In their work, Florou et al. also considered supporting or attacking relations.

The second sub-task, which involves the classification of argument components, has been addressed in early works such as [15], which focused on scientific articles. In this work, each sentence is classified into one of seven rhetorical categories based on its argumentation scheme: background, topic, related-work, purpose, method, result, and claim. These categories are based on the sentence's role in the article, with a focus on results and features. Similarly, [16] annotated the Araucaria corpus using a non-domain-specific scheme to identify claims, premises, and non-argumentative text units based on specific features, and achieved notable results. Additionally, [17] addressed the subtask of identifying argumentation schemes on the Araucaria corpus by applying the principles of argumentation theory from [14].

Regarding the third sub-task, which involves the identification of argumentation structures, previous works have focused on different domains.[18] and [19] have addressed this task in the legal domain, where argumentation is well-structured and often leads axiomatically to conclusions. They identified argumentation structures as patterns that rely on discourse markers. In contrast, [20] proposed an annotation scheme for general texts consisting of three elements: major claim (a main statement for the whole text), claim (minor statements), and premise (basis for the claims and major-claims). The argumentation structure is defined by directed relations between the elements, including premise → premise, premise → (major-) claim, and claim → major-claim. These relations can be either *supporting* or *attacking*.

In a similar approach, [21] annotated a corpus of 25,000 instances from controversial topics (such as abortion, climate change, nuclear energy, etc.) at the sentence level using three labels: *argument-pro*, *argument-against* and *no-argument*. This annotation scheme is based on the definition of an argument as a span of text expressing evidence or reasoning used to support or attack a specific *topic*. In terms of previous works, the topic is equivalent to a *(major) -claim*, and *arguments* would be equivalent to *evidences* with two possible relations (*support* or *attack*), which are encoded in their label. Compared to previous annotation schemes, this approach has several advantages. First, it is applicable to the task of argument search and is general enough to be used with a variety of data sources. Additionally, it is simpler to understand for untrained annotators.

Moreover, we have found that a simplified approach, where the three sub-tasks are condensed into a single one, is the most suitable for our problem. Our dataset consists of sets of reviews as separate documents, all referring to a common *topic*: the use of a drug for a specific medical condition (a *use case*, in our terms). Identifying text units as *argument_for*, *argument_against* or *non-argument* instead of identifying *claims* and *premises* (which are not so clearly distinguishable in informal texts such as reviews) and their relations reduces significantly the complexity of the problem and the annotation. Furthermore, in the vast majority of cases, every piece of argumentative information is always directed towards the topic and not to other pieces of information. Additionally, we are interested in aggregating all these arguments in future tasks to extract an objective view from all this opinionated content, which makes our problem closer to the task defined in [21] as *argument search*.

Following a similar approach and using the annotation schema defined by [21] (*supporting argument*, *attacking argument*, *non related*), we have created a novel dataset for argument detection on drug reviews from online forums, named NAME. The dataset consists of 6,040 sentences in 858 user reviews from six different *topics* (tuples consisting of drug and health condition) chosen to represent common, unrelated, and diverse medical conditions (Bowel-Preparation, Birth-Control, Diabetes-Type-2, Acne, Depression, and Influenza). The dataset is based on the original *Drug Review Dataset* presented in [2]. Six trained annotators labeled each sentence as either *supporting argument*, *attacking argument*, or *non related*.

We tested various automatic classification methods, including classical machine learning models, transformer-based models, and ensembles of both types, for binary (*argument* vs. *no-argument*) and ternary (*non argument*, *supporting argument*, *attacking argument*) classification problems. Our results indicate that this task can be successfully approached using these methods, and the performance is consistent across different types of classification systems.

The feasibility of achieving high performance in classification systems for this task lays the foundation to accomplish further goals and tasks, including detecting controversy, mining facts and side effects, among others. Combining argument detection in drug reviews with existing work on sentiment analysis can provide a more comprehensive understanding of users' experiences with drugs.

3.1.1. [TASK: Re-locate] - Older state of the art state-of-the-art

A good overview and description of the process can be found in [22], performed on the argument mining corpus [23] and persuasive essay's corpus considered in [10]; a series of *features* proposed by the work of [10] consisting of different types: structural, lexical, syntactic, argumentation indicators, contextual, *word-embeddings*, etc. A binary classification task is performed at the argument subcomponent level (*evidence*, *claim*, *none*), using different classification models: Logistic Regression, *Support Vector Machine*, *Random Forest*, *Naïve-Bayes*, *Adaboost*, *Convolutional Neural Networks*, etc. achieving results of $F1 = .66$ with the for a set of 3832 persuasive [10] trials and $F1 = .94$ for the case of 2858 data instances of *Wikipedia* [23].

[TASK: include newer references :: Gurevych has at least 1 more paper on this problem]

Such difference in the performance suggests that the problem is highly domain-dependent: free and more personal style in persuasive essays versus common structures and language in *Wikipedia* articles.

[TASK: decide whether this should be here or in the Dataset Section]

Following this line, we find the approach of [21], in which a corpus of our interest is presented, consisting of 25,000 instances annotated at sentence level as (*argument-pro*, *argument-against*,*no-argument*) for 8 different topics considered controversial. In this case, no *features* engineering has been performed, but a 300-dimensional *word-embedding* is used from *word2vec* [24] to represent as vectors the instances and various versions and configurations of Long-Short-Term-Memory type [25] neural networks are used achieving as $F1 = 66.62\%$ in the best case for binary classification and a $F1 = 42.85\%$ for the described three-category classification.

3.1.2. WORK IN PROGRESS: Re-locate - Recent state-of-the-art

* also found as argument mining and argument retrieval

Surveys. (Cabrio and Villata 2018) Five years of argument mining... [26] “Argument mining is the research area aiming at ex- tracting natural language arguments and their re- lations from text, with the final goal of providing machine-processable structured data for computa- tional models of argument.”

(Lawrence and Reed) “Argument mining is the automatic identification and extraction of the structure of inference and reasoning expressed as arguments presented in natural language. Understanding argumentative structure makes it possible to determine not only what positions people are adopting, but also why they hold the opinions they do, providing valuable insights in domains as diverse as financial market prediction and public relations” [27]

Some domains have been particularly broadly studied for the task of argument mining. This is the case of *persuasive online discussions* [28] (applications of transfer learning, based on *Reddit* threads data. Another social network that provides a solid source of data is *Twitter* [29] (first corpus of tweets annotated at level of sub-tweet, with similar schema and methodology) [30]

Policy domain [31];

Legal collections [32, 33, 34, 11, 19](Mochales and Ieven, 2009, Savelka and Ashley, 2016, Yamada et al. 2017, Moens et al. 2007, Wyner et al. 2010); [35]

Student essays ([21, 36]Stab, Miller, Gureych 2018, Persing and Ng 2016);

User comments on proposed regulations (Park and Cardie 2014); [37]

Newspapers articles (Feng and Hirst 2011) [17]

Political debate fora (Walker et al. 2012, Abbott et al. 2016); [38, 39, 40]

Persuasive Essays (Stab and Gurevych 2014, Carlile et al. 2018);[10, 41, 42, 43]

Biomedical publications [44, 45, 46, 47](Wilbur 2006, Blake 2010, Achakulvisut 2019, Mayer 2020);

Newspapers, blogs and social web (Goudas 2015, Kiesel 2015, Habernal and Gurevych 2017). [48, 49, 50]

Argument retrieval (Rastegar-Mojarad 2016) [51]

We can find several works of argument mining in the medical domain, particularly focused in clinical trial data ([52]). In this case, they use again the framework presented by [20] where the argumentation schema consists of *major claim*, *claim* and *premise* and the argumentation process consisting of three steps ((i) identifying argument components, (ii) classification of argument components – the schema previously mentioned —, (iii) identification of argument relations, (iv) structure representation, and (v) stance recognition. Similar works are [53], also based on clinical trials texts and applying novel techniques (sequential multi-tasking learning). Similar work in [54] , a dataset on clinical trials (Randomized Controlled Trials) was developed using BERT, LSTM, GRU and CRF on a similar schema of two classes (*evidence* and *claims*) and two relations (*support* and *attack*). Also in the context of health and policy (COVID-19), there are works on tweets [55] Additional research focusing on COVID-19: Workshop COLING 2022 - classification of stance and premise in tweets about COVID-19 health mandates; reviews in WebMD; tweets with self-reported COVID-10 systems, vaccination status, classification of self-reported chronic stress, disease mentions, intimate partner violence,

Additional techniques have been proposed and benchmarked in [56] for different datasets including user-generated comments (Cornell eRulemaking Corpus consisting of user comments on about the consumer rights in receiving financial services) , scientific publications, and persuasive essays. In [57] it is shown that using multiple corpora to train a multi-task learning approach is useful to improve overall performance for specific datasets.

Among the state-of-the-art techniques, it seems that the use of *attentive residual networks* is the one of the best performing architectures tested cross-domain.

Another domain where argument mining has gained a lot of attention is the legal domain [58], [59], [60]

BERT-based transformers are widely used for this task in different configurations: self-attention based embeddings [61] using the CDCP dataset previously mentioned.

* The problem of annotated data scarcity has forced the proposal of different cross-domain approaches [62], where knowledge transfer has been shown to be useful...

Systems [63] ways to design systems that use large datasets and annotation mehtodologies...

3. By technology: transformers, LLMs?

4. Tasks: classification, KG generation,

Alternative methodologies to address these tasks is Zoning

In our case, the format of the documents and the domain allows us to define a central *major claim* that it is implicit in the text, but the trigger of the whole text production: a drug is used for a condition. Following the steps of literature, it seems that we need to apply, at least, two out of the four steps (detection, relations, structure, etc.), which are detect the class (*claim*, *evidence*) and the relation (*support* and *attack*). Given the granularity of the conversations, we adapted the broadly-used schema from previous works initially proposed in [20] Probably, a qualitative difference to our case is the free text generation of a user in a very defined use case where the standard approach is to expressed pro- and contra- items, alongside with a short introduction and conclusion (optionally). We can consider a two-level schema where in a upper level we find the main class *major claim*, fixed and pre-defined, which is the target of all argumentation; and the second level (actually, the classes to annotate) that are *supporting* and *attacking*. In this definition, the relations are encoded in the argumentation schema directly as classes. This allows us to execute For consistency reasons, we need an additional class called *non_related* to address all those text snippets that have different nature to the

3.2. Corpora for Argument Extraction

One noteworthy dataset developed for argument mining tasks is the corpus presented in *Aharoni et al.* [23]. This dataset comprises hundreds of articles extracted from *Wikipedia* and includes 2,683 annotated argumentative elements across 33 different controversial topics. The definition of an **argument** in this dataset consists of two elements, a **claim** and **evidence**, which do not necessarily need to constitute complete sentences. The **topic** is a short sentence defining the subject of interest. A *Context Dependent Claim (CDC)* is a general sentence that either supports or contradicts the *topic*, while a *Context Dependent Evidence (CDE)* is a text segment that directly supports the *CDC* within the context of the *topic*. Evidences can be further classified based on their basis, such as *study*, *expert*, or *anecdotal*.

Another interesting dataset is the one developed by *Stab et al.* [20]. This dataset introduces a new approach for modeling arguments, their components, and relationships in persuasive essays written in English. The annotation scheme involves separately annotating *claims* and *premises*, as well as identifying *support* or *discredit* relationships. In this dataset, an argument is defined as a textual structure consisting of various components, primarily *claims* (exposition of an idea) and *premises* (elements that support or discredit a claim). Based on this definition, we can identify several subtasks involved in recognizing arguments in texts:

1. Separating *argumentative* from *non-argumentative* text units.
2. Identifying *claims* and *premises*.
3. Identifying relationships among the components of argumentation, such as *supports*, *attacks*, etc.

The novelty of this corpus lies in its focus on persuasive texts. Although these types of texts have been studied previously in other contexts, such as in the area of *Automatic Evaluation of Essays*, which aims to automatically assign a grade to student essays based on certain features, this is the first attempt to use persuasive texts for argument mining.

Regarding the proposed scheme, its goal is to model argument components as well as argument relations, so the scheme shown in Figure 1 is proposed.

The study of persuasive essays reveals a common structure: usually the introduction includes a *major claim* that expresses the author's general position on the subject. This *major claim* is supported or confronted by the arguments covering certain aspects in the following paragraphs. Some examples of *major claim* may be (marked in bold):

- I believe that **we should attach more importance to cooperation during education**.

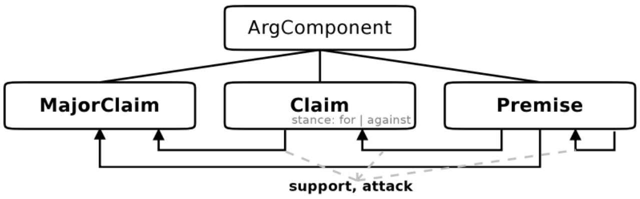


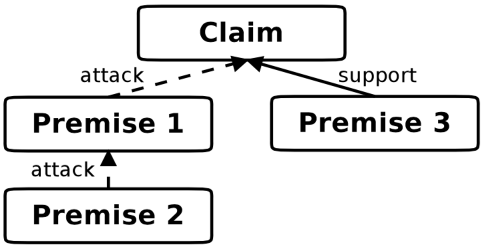
Figure 1: Schema annotation of arguments, their components and relationships

- From my viewpoint, **people should perceive the value of museums in enhancing their own knowledge**.
- Whatever the definition is, **camping is an experience that should be tried by everyone.**

As for the relations between argumentation components, only two relations are distinguished: *support* and *attack*. Both relations can occur either between two *premises*, a *premise* and a *(major-) claim* or between a *claim* and a *major-) claim*.

We can visualize this through the following example, depicted in Fig.2:

- **Living and studying overseas is an irreplaceable experience when it comes to learn standing on your own feet**. One who is living overseas will of course struggle with loneliness, living away from family and friends$_1$ but those difficulties will turn into valuable experiences in the following steps of life$_2$. Moreover, the one will learn living without depending on anyone else $_3$



.

Figure 2: structure diagram of the example structure.

We can observe how *claim* is attacked by *premise* $_1$, while *premise* $_2$ is a refutation of *premise* $_1$. In turn, *premise* $_3$ directly supports *claim*.

Ultimately, the corpus consists of 90 persuasive essays, selected from *essay forum*, a forum with an active community that contributes written feedback on different types of texts. The final corpus includes 1,673 sentences with 34,917 tokens.

Another relevant dataset is introduced in [21], which is constructed from multiple heterogeneous and controversial sources. This dataset presents a novel sentence annotation methodology that can be applied to various domains. This approach effectively mitigates the domain dependency issues observed in previous studies, making it a valuable contribution to the field. In other prior studies, the focus has been on identifying argumentation structures in texts that possess specific characteristics and pertain to particular topics, such as texts from *Wikipedia*. However, the effectiveness of these approaches when applied to other text types and domains is questionable, and adapting them would necessitate significant efforts.

To address these challenges, [21] proposes the following solutions:

1. A novel scheme that focuses on the information search perspective in argument search. This scheme is designed to be sufficiently generalist to be used on a large heterogeneity of data sources, and simple enough to be applied manually by untrained annotators.

2. An annotated corpus that covers eight controversial topics, comprising over 25,000 instances. This corpus stands as the first known resource that enables the evaluation of cross-topic Argument Mining methods using heterogeneous sources.
3. An exploration of different approaches to incorporate information into neural networks and enhance cross-topic learning tasks.

In [21], an *argument* is defined as a text fragment that expresses evidence or reasoning and can be utilized to support or oppose a specific topic. While an argument may rely on information from its context, it should not be ambiguous in terms of its orientation towards the topic. On the other hand, a *topic* refers to a subject of controversy that allows for clear polarity, typically resulting in instances that are either *in favor* or *against* the topic.

We can observe these criteria in the examples provided in Fig.**??**. It is worth noting the instances that have been classified as non-argumentative since simply taking a stance *in favor* or *against* a topic is insufficient. To be considered argumentative, a statement must also include reasoning or evidence to support its position.

The data for the corpora was manually collected, encompassing a diverse range of topics and text types. Eight topics were extracted from various online lists of controversial teasers. For each topic, Google searches were conducted, and non-archived results were excluded using the *Wayback Machine*. The resulting list of search results was then truncated to 50 entries.

The resulting dataset comprises web documents that contain controversial content, including documents from newscasts, editorials, blogs, discussion forums, and encyclopedia articles. The text underwent standard preprocessing procedures, and further filtering was applied by removing sentences with fewer than four tokens or lacking verbs. This filtering resulted in a total of 27,520 sentences.

To annotate the dataset, each sentence was classified as either *pro-argument*, *counter-argument*, or *non-argument*. The category of *non-argument* includes sentences that are ambiguous, incomprehensible, or fall into less than 1% of cases. The annotation process involved two teams: one comprised of expert annotators, and the other composed of previously untrained annotators. Both teams classified the same set of sentences to reduce bias.

Consequently, each sentence was independently annotated by two experts and ten inexperienced annotators. The annotations were then compared, and if the level of inter-annotator agreement surpassed 0.721 for Cohen's κ [64], the annotation was considered reliable, and a *gold-standard* was established.

topic	sentence	label
nuclear energy	Nuclear fission is the process that is used in nuclear reactors to produce high amount of energy using element called uranium.	non-argument
nuclear energy	It has been determined that the amount of greenhouse gases have decreased by almost half because of the prevalence in the utilization of nuclear power.	supporting argument
minimum wage	A 2014 study [. . .] found that minimum wage workers are more likely to report poor health, suffer from chronic diseases, and be unable to afford balanced meals.	opposing argument
minimum wage	We should abolish all Federal wage standards and allow states and localities to set their own minimums.	non-argument

annotation examples for the corpus, at sentence level for different domains.

4. The DrugArgDet Dataset

As outlined in Section 1, our objective is to develop a system for identifying arguments within drug reviews from online forums. These arguments hold significant value as they often contain valuable and fact-based information regarding real experiences with drugs. By detecting these arguments, we can filter out irrelevant text segments that are less likely to contain relevant information. This, in turn, paves the way for addressing further objectives and challenges, including the identification of controversial drug use cases, mining side effects, generating statistics on actual drug usage and acceptance, and extracting information that currently relies solely on patient reports and monitoring by healthcare practitioners and organizations.

To accomplish our goal, the first step is to locate or create a suitable dataset within this domain that can be used for training and evaluating our argument detection system. To identify such a dataset we must consider three criteria: *domain* (drugs and health), *type of document* (user reviews), and *existing labeling* for the task (argument detection).

Regarding the domain, we came across a potential candidate in the form of drug reviews provided by users in online healthcare forums ([2]). This dataset aligns with our domain and the desired type of document. However, it lacks the necessary labeling for argument detection. Also within the health domain, there are several datasets available that focus on clinical texts annotated with argumentation schemes ([47, 54, 65]). These clinical texts typically exhibit structured patterns with clear entities, such as *patient with <condition> and <dose> <drug> was prescribed, resulting in <results>*. Additionally, they often employ medical formal language and feature well-structured argumentation. However, this differs from the usual informal language and less structured nature of user reviews. Consequently, these clinical text datasets do not fulfill the criterion of the desired document type.

Regarding the *type of document*, it is worth noting that user reviews for certain products can exhibit common characteristics such as a similar tone, structure, and language. However, it is important to recognize that user reviews on non-medical products or experiences may lack the underlying factual information. These reviews often tend to be subjective, driven primarily by personal taste and opinion [6, 66].

In terms of datasets with *existing labeling*, it is important to note that different approaches have been taken in various cases for argument detection. These approaches often rely on domain-specific features, making them less generalizable. Examples of such datasets include the *Persuasive Essays* corpus [10], a *Wikipedia corpus* [23], a *Twitter corpus*, the *Araucaria corpus* [23], and others that cover diverse text types such as *microtexts* [67], *scientific articles* [68], *various* text sources [69], and *controversial topics* [21], among others. None of these datasets fulfill the criteria of both *type of document* and *domain* that are relevant for our study.

After carefully considering the available options, it is evident that a specialized dataset focused on drug reviews, labeled with argumentation schemes, has not been developed thus far. Consequently, our primary objective for this research is to create such a dataset. Our approach involves labeling the Drug Review Dataset[2] using an appropriate argument detection scheme, drawing inspiration from previous works on argumentation detection [21].

The Drug Review Dataset[2] comprises a comprehensive collection of drug reviews sourced from popular portals such as Drugs.com and Druglib.com. Users submit their reviews along with details about their medical conditions and a rating based on their personal experiences. This dataset comprises a total of 218,614 reviews for various drugs and medical conditions. To facilitate training and evaluation, the dataset has been divided into two stratified partitions: 75% for training and 25% for testing purposes. Previous studies have extensively explored this dataset, primarily focusing on sentiment analysis by utilizing the rating as a label. Additionally, researchers have delved into opinion mining, specifically examining the underlying semantics of the reviews.

For our research, we aim to leverage a suitable annotation scheme, as described in [21]. This scheme involves three distinct classes: *irrelevant*, *supporting argument*, and *attacking argument*. By employing this annotation scheme, we can effectively analyze the argumentation present within the drug reviews.

To enhance the annotation scheme, we further define the concept of the *major claim* to which our supporting and attacking arguments are directed. We assume that behind each review lies an implicit *topic*, specifically expressed as *”the use of <drug> is recommended for <medical condition>”*. This viewpoint arises because users provide their reviews based on their experiences with the drug in relation to their medical condition. Consequently, we consider all relevant arguments, whether *supporting* or *attacking*, to be directed towards this *major claim* (see Figure **??**). Similar to the approach in [21], we adopt the sentence as the unit of annotation. This decision constitutes a good *trade-off* between the potential for information extraction and the complexity of annotation. Given that the texts in the dataset are informal, they often lack structure, exhibit incongruent grammar, employ colloquial language, and lack clause-based argumentation.

Typically, a review encompasses several types of information, including personal context, information regarding the intake of the drug, including arguments for or against its use, sentimental or subjective expressions (”I hate it!”, ”I love it!”, ”This makes my life better”), and references to other drugs, often making comparisons with the drug being reviewed (”This is much better than <another drug>”, ”I will never go back to <another drug> again”). Among these elements, relevant factual information regarding the user’s experience lies within the text units that express arguments in favor or against the drug. Consequently, we consider personal and subjective information *non relevant* for the purpose of this task.

We define the combination of a specific drug and medical condition as a *use-case*, which forms the foundation of

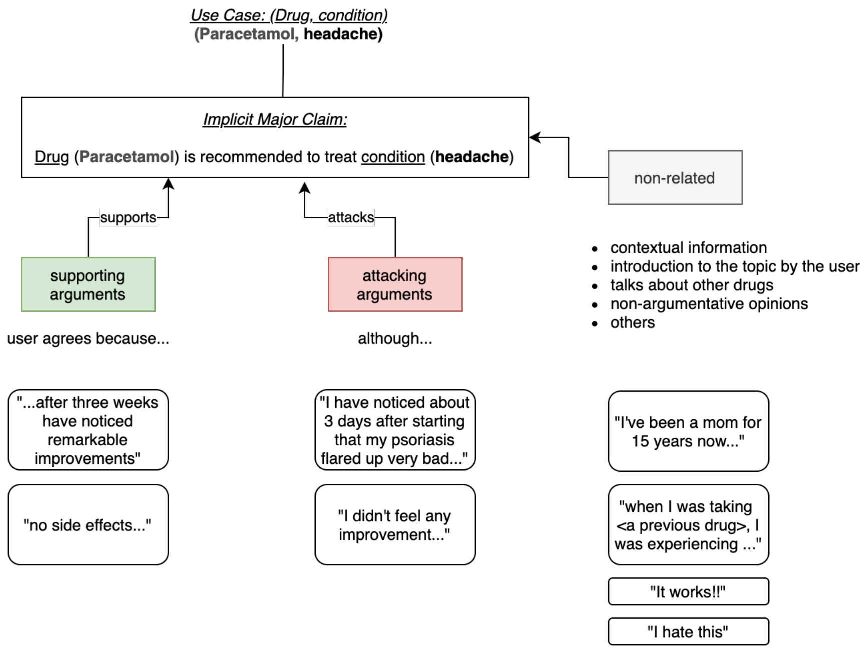


Figure 3: Every sentence is considered in the context of its review. The annotator should consider the overall theme of the review, as well as the preceding and subsequent sentences. The same sentence may belong to different categories depending on the content of the previous or following sentences. Thus, the entire review will be displayed, with sentences divided, and the annotator labels them individually. Each sentence is assigned only one label, which can be either **supporting**, **attacking**, or **non related**.

our defined *major claim* ("The use of <drug> is recommended for <medical condition>"). Upon conducting an initial manual review of a randomly selected sample, we have observed slight variations in textual and linguistic properties across different *use-cases*. As a result, it is prudent to identify a set of representative *use-cases* for further analysis and study.

We have chosen six representative *use-cases* from the training split of the dataset presented in [2]:

- The use of *Suprep* for *bowel preparation*.
- The use of *Ethynil-stradiol* for *birth control*.
- The use of *Liraglutide* for *diabetes*.
- The use of *Ethynil-stradiol* for *acne*.
- The use of *Prozac* for *depression*.
- The use of *Oseltavimir* for *influenza*.

To ensure diversity, we have avoided selecting multiple cases for closely related medical conditions, such as anxiety and depression, or acne and facial skin problems. Table 1 provides further details on this selection process. Furthermore, we have intentionally selected two *use-cases* for the same drug, Ethinyl-estradiol-/-norgestimate, to facilitate a contrastive analysis. This particular drug is highly prevalent in the original dataset and is prescribed for distinct *use-cases* like birth control and acne treatment.

For each chosen *use-case*, we have selected approximately one thousand sentences, preserving the original review structure without dividing it. This approach enables annotators to label the sentences directly within the review text, enabling them to grasp the contextual nuances. To enhance clarity during the annotation process, the annotators have visibility of the *major-claim* at the top of the annotation window (see Figure 4).

The annotation process involved six annotators divided into two teams, with each team annotating three use-cases to minimize individual biases during annotation. The final label was determined by a majority vote. In cases where

use_case	# docs	# text_units
(Suprep-Bowel-Prep-Kit; Bowel-Preparation)	110	1003
(Ethinyl-estradiol-/-norgestimate; Birth-Control)	160	1038
(Liraglutide; Diabetes,-Type-2)	139	987
(Ethinyl-estradiol-/-norgestimate; Acne)	140	988
(Prozac; Depression)	159	1005
(Oseltamivir; Influenza)	150	1019

Table 1: Use cases defined by the tuple (drug, medical condition), with number of reviews (docs) and number of sentences (text units) per use case. We have extracted approximately one thousand sentences per use case, preserving the documents undivided.

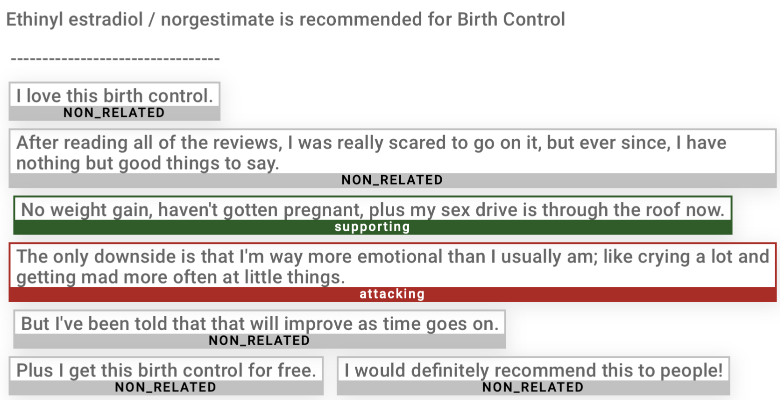


Figure 4: View of annotation process for one review. At the top, the annotator can read the *major-claim* and select the desired label for each one of the sentences in the review (*NON_RELATED*, *supporting*, *attacking*)

there was a complete disagreement, an additional annotator from the team not involved in that specific annotation was consulted (which accounted for only 0.05% of the total annotations). All annotators underwent extensive training using an annotation guide before the process began. The annotation process was conducted in three stages, with each stage comprising approximately 33% of the total annotations. The annotation tool *Doccano*[70] was used for the annotation process. Furthermore, annotators were instructed to provide comments on ambiguous cases, which were discussed in joint meetings after each stage. The annotation guidelines were slightly adjusted if necessary during these meetings. After completing the entire annotation process, the annotators had the opportunity to review their annotations once more, taking into account the revised and more consolidated guidelines.

We evaluated the inter-annotator agreement using Cohen's kappa coefficient (McHugh, 2012), and the average value obtained was 0.72 (see Table 2). A low value of kappa would indicate an ambiguous annotation process or excessive difficulty for the annotators. However, values above 0.6 are considered indicative of substantial agreement and are considered suitable for this type of task [71].

	Kappa
Annotators 1-2	0.720270
Annotators 1-3	0.721038
Annotators 2-3	0.717789
Average	0.719699

Table 2: Inter-annotator agreement using Cohen's Kappa as metric. Values between 0.6 and 0.8 are considered as substantial agreement, which is adequate for this kind of task.

Upon reviewing the final label counts, it is evident that the ratios for the three classes are remarkably similar (see

Figure 5). Furthermore, it is worth noting that approximately two-thirds of the sentences are labeled as *non related*, which aligns with our expectations.

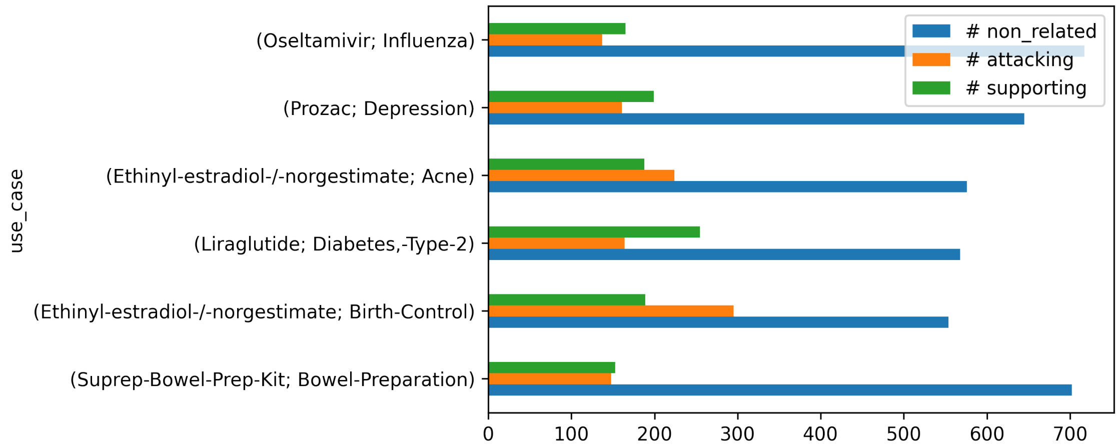


Figure 5: Distribution of labels per use-case.

Finally, Figure 6 shows the most frequent terms for each use case. We may extract the following conclusion from these statistics regarding each use case:

- Suprep for bowel preparation: The conversations usually revolve around not only the drug itself but also the application device and the procedure, as well as the results. The presence of medical concepts is surprisingly low.
- Ethynil-stradiol for birth Control: This use case features mentions of long-term therapies. There is a strong presence of concepts regarding side effects of the drug as well as other terms that refers to emotions or behavioral aspects (such as love or mood).
- Liraglutide for diabetes: Entities related to quantities are very relevant, reflecting users' concerns about dosage. Mentions of side effects are alsoo notable.
- Ethynil-stradiol for acne: This use case has high references to long-term timings indicating that descriptions of long-term therapies are likely. The condition was frequently mentioned, possibly because the drug serves multiple purposes, such as birth control, cystic ovarian and acne treatment. Side effects are also mentioned.
- Prozac for depression: The most relevant aspect in this case is the mentioning of other people, along with a relatively high presence of quantity entities. This relates to experiences involving third persons, such as relatives, and different drug dosages used in treatments. Concepts related to mental health conditions and processes are frequent.
- Oseltavimir for influenza: Frequent entities in this case are related with time, reflecting the fact that the condition evolves quickly.There is a noticeable presence of symptoms.

5. Automatically Detecting Arguments in Drug Reviews

The objective of this section is to present various approaches to argument detection, including classical machine learning techniques as well as newer transformer-based approaches. These approaches have been tested to evaluate the suitability of our dataset and to assess the feasibility of automatically detecting and classifying arguments in drug reviews.

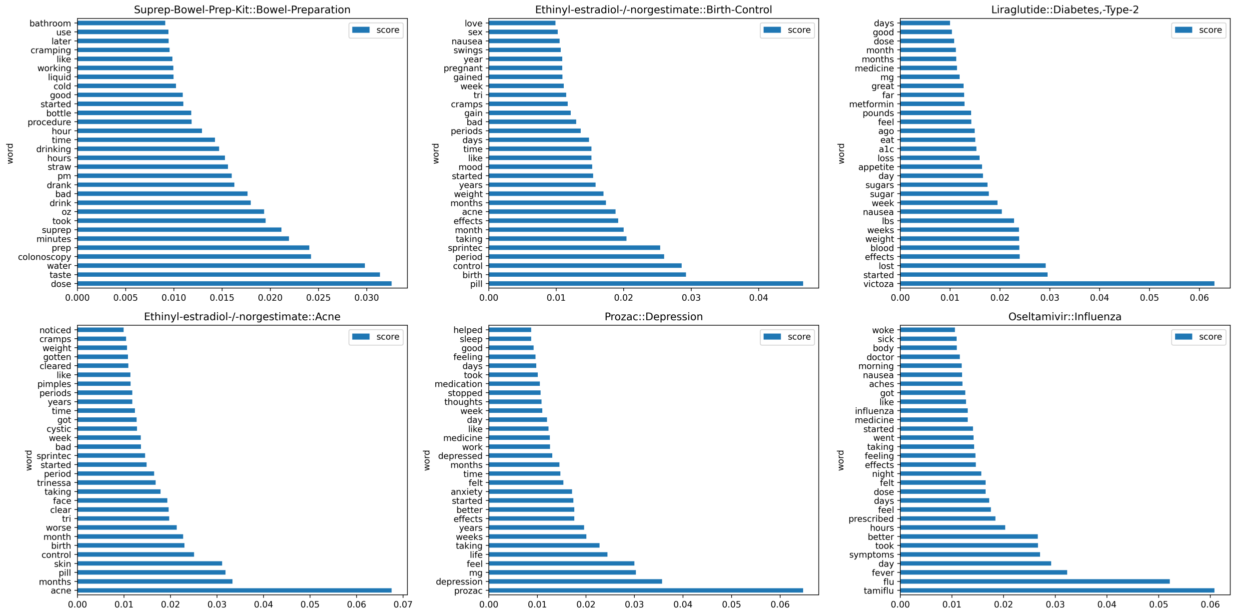


Figure 6: Top 30 terms per use case (tf-idf)

As described in Section 1, the problem of argument detection is usually approached in the literature as a sentence-level classification problem with three labels: *non related*, *supporting*, and *attacking*. Additionally, in our experimentation we have also considered a binary classification task where we combine *attacking* and *supporting* as *related* (i.e., argumentative towards the *major claim*), and *non related*.

5.1. Classical Machine Learning Approaches

We have explored a set of classical machine learning models, which are readily available in the *scikit-learn* library [72][1]. These models are widely used in text classification research, including Support Vector Machine (SVM) [73], Logistic Regression (LR), Random Forests (RF), Stochastic Gradient Descent (SGD) Classification [74], and XGBoost [75].

To extract features from the text, we first applied a combination of text preprocessing techniques, including tokenization, stopword removal, stemming, and lemmatization. In order to maintain fairness in the comparison between models, we employed an identical set of features across all models.

Following [20], we compute and test the following features:

- Term Frequency-Inverse Document Frequency (TF-IDF): we convert the preprocessed text into numerical representations suitable for machine learning models.
- Syntactic features, including the POS distribution (how many token of each POS type are in the sentence), the number of subclauses in sentence, the depth of constituent parse tree of the sentence, the tense of the main verb and a boolean indicating the presence of modal verbs.
- Text statistics, such as the length of the post containing the sentence and the number of sentences.
- Lexical indicators, including ”forward indicators”, which indicate that the component following the indicator is a result of preceding argument components.(e.g., “therefore, “thus”, or “consequently”); ”backward indicators”, which indicate that the component following the indicator supports a preceding component (e.g., “in

[1] `https://scikit-learn.org/`

addition", "because", or "additionally"); "thesis indicators", which point to major claims (e.g., "in my opinion" or "I believe that"); and "rebuttal indicators", which signal attacking premises or contra arguments (e.g., "even though", "however", "otherwise").

- Sentiment, a binary feature indicating whether the sentence entails "positive" or "negative" emotions. This feature is computed using the bert-base-multilingual-uncased-sentiment model from Huggingface [2].
- Embeddings, word vectors of 400 dimensions computed using the GloVe algorithm (Global Vectors for Word Representation) [3].
- Medical feature, a number indicating the frequency of medical concepts in the sentence, detected using the MetaMap 2018 semantic types[4].

5.2. Transformer-Based Approaches

Our second set of approaches involves the use of Transformer-based techniques, particularly those that rely on BERT (Bidirectional Encoder Representations from Transformers) [76]. Specifically, we employed RoBERTa [77] and BioBERT [78]. RoBERTa is an optimized variant of BERT, designed for enhanced performance, while BioBERT is a domain-specific adaptation of BERT, pretrained on a substantial corpus of biomedical text.

We employed the Hugging Face Transformers library[5] to fine-tune both RoBERTa and BioBERT for our classification tasks. The models were initialized with their pretrained weights, and a classification head tailored to the specific task was added. During training, we utilized a batch size of 16, a learning rate of 2e-5, and a maximum sequence length of 512 tokens. The transformer models were fine-tuned for a total of 3 epochs, with early stopping based on monitoring the validation loss.

6. Results and Discussion

This section introduces the findings of the study and discusses such findings in the context of our research questions. First, we present the evaluation framework. Second, we summarize the results of our evaluation. Finally, we discuss the findings and draw conclusions.

6.1. Evaluation Methodology

In our experiments, we utilized a 10-fold cross-validation approach to train and test our models. Each fold was divided into a training set (80%) and a testing set (20%), and the training set was further partitioned into a training set and a validation set, with the validation set constituting 20% of the original training set. To ensure a fair and representative distribution of the target variable (label), we applied stratified division during the dataset splitting process.

We conducted hyper-parameter tuning through grid search and cross-validation on the training set to determine the best model configurations. The models' performance was evaluated on the testing set.

Table 3 shows the different feature configurations tested for the traditional approaches for which the results are presented. Other combinations were tested but drawn worse results, and thus they are not presented. Moreover, for each configuration, only the results for the best ML algorithm are presented

To ensure the stability and reliability of our results, we conducted multiple experiments using different random seeds. Next, we performed a comprehensive analysis of the performance metrics to gain insights into the strengths and weaknesses of each model within the context of our NLP classification tasks.

It is worth noting that although the classification unit was the sentence, we preserved comments (containing one or several sentences) to maintain annotation consistency and prevent data leakage. Despite the class imbalance in our

[2]https://huggingface.co/nlptown/bert-base-multilingual-uncased-sentiment
[3]https://nlp.stanford.edu/projects/glove/
[4]https://lhncbc.nlm.nih.gov/ii/tools/MetaMap/documentation/UsingMetaMap.html
[5]`http://huggingface.co`

Configuration	TF-IDF	Text stats	Syntactic	Indicators	Medical	Embeddings	Sentiment
1	✓						
2	✓	✓					
3	✓	✓	✓				
4	✓	✓	✓	✓			
7	✓	✓	✓	✓	✓		
10	✓	✓	✓	✓	✓	✓	
13	✓	✓	✓	✓	✓	✓	✓

Table 3: Different experimental configurations for the classic machine learning methods.

dataset (60%, 40%) for the binary scheme and (60%, 20%, 20%) for the ternary scheme, we chose not to extract a balanced sample, opting instead to retain the original distribution.

Our models' performance was evaluated using the F1 score as the primary performance metric for both the binary (argument, non-argument) and ternary (supporting-argument, attacking-argument, non-related) classification schemes. This allowed us to gauge our models' effectiveness in handling imbalanced class distributions and to assess their overall performance on the given NLP classification tasks.

6.2. *Results*

[bullet points to be written as paragraph]

Table 4 shows the results for the top performing configurations of both the traditional machine learning models and for transformers, and for both the 2-classes and 3-classes argument detection tasks.

Conf.	Algorithm	2-classes task			3-classes task		
		P	R	F1	P	R	F1
1	Random Forest	0,75	0,76	0,75	0,56	0,63	0,58
2	Random Forest	0,75	0,76	0,76	0,55	0,63	0,57
3	SGD Classifier	0,74	0,75	0,75	0,59	0,62	0,60
4	SGD Classifier	0,74	0,75	0,75	0,58	0,62	0,60
7	SGD Classifier	0,75	0,75	0,75	0,59	0,62	0,60
10	SGD Classifier	0,75	0,75	0,75	0,59	0,62	0,60
13	SGD Classifier	0,75	0,75	0,75	0,60	0,63	0,61
-	Distilbert base-uncased	0,81	0,81	0,81	0,75	0,75	0,75
-	Bio ClinicalBERT	0,82	0,83	0,82	0,76	0,76	0,76

Table 4: Selection of the top performing configurations for both traditional machine learning models and for transformers.

We first focus on analyzing the results of the traditional approach and the capability of the different features employed. We observe that the SGD algorithm provides the best results in nearly all the experiments, regardless of the features employed EZEQUIEL - Por qué crees que puede ser? QUé algoritmo lleva por debajo la implementación que has utilizado? es stochastic gradient descent, estoy intentando encontrar una respuesta a por qué puede ser esto así. Una cosa que hay que mencionar es que, si miras a los resultados, parece que llegamos a algún tope en el 0.75, da igual las features que añadamos de más, pero lo cierto es que hay que mirar a mirar con la granularidad de los casos de uso – que menciono más abajo – porque al final la media es .75, pero hay casos de .80 y casos de .64. Y lo más difícil es evidentemente depresión

We have observed that, in the binary classification task, utilizing TF-IDF alone yields satisfactory classification results, and the inclusion of additional features does not lead to significant improvements. However, the incorporation of the text statistics feature shows a slight enhancement in the F-1 score. Conversely, in the three-class classification task, considering all the features leads to improved F-1 scores.

As expected, the results for the binary classification (argument, no-argument) outperform those of the ternary classification (supporting-argument, attacking-argument, non-related) in every configuration and evaluation metric.

However, it is worth noting that the ternary classification scheme presents a more nuanced and detailed approach to analyzing the text, as it allows for a distinction between different types of argumentative relationships. This can be particularly useful in specific contexts where a more fine-grained analysis is required.

Our results show that transformer-based approaches outperform classical approaches in both binary and ternary classification tasks. Specifically, the best-performing transformer-based model achieved an F1 score of 0.82 for the binary task, while the best-performing classical model (a combination of SVM classifier and unigram and bigram features) achieved an F1 score of 0.76 for the binary classification task. For the ternary classification task, the best-performing transformer-based model achieved an F1 score of 0.76, while the best-performing classical model achieved an F1 score of 0.68. The differences in performance between the best-performing models of both families were significant, with a $\Delta F1_{binary} = 6\%$ and $\Delta F1_{ternary} = 11\%$. Therefore, transformers seem present additional advantage in the most complex ternary classification task.

Notably, the performance of one of the simplest configurations, a Random Forest classifier using TF-IDF features, achieved a surprisingly high F1 score of 0.75, especially when compared to the best-performing transformer-based model in the binary classification task. This highlights the potential of simpler models for certain NLP classification tasks, where computational efficiency and interpretability may be of greater importance than state-of-the-art performance. However, it is important to note that the best-performing transformer-based models still outperformed the Random Forest classifier in both binary and ternary classification tasks, indicating the ongoing superiority of transformer-based approaches in NLP.

Despite the good performance of classical approaches, our analysis revealed that both types of approaches appear to be detecting different aspects of argumentation. Specifically, when examining the well- and wrong-predicted instances from both types of models at the level of individual use cases, it became apparent that each type of model had different strengths and weaknesses. Therefore, given the differences across use cases, it seems advisable to analyze results at level of granularity of use case.

Concerning the binary classification task (argumentative vs. non-argumentative), our analysis revealed that classical models displayed dissimilar behavior across different use cases. Specifically, we observed in Table 5 three levels of performance, with the cases Liraglutide::Diabetes-Type-2 and Ethinyl-estradiol-/-norgestimate::Birth-Control exhibiting the highest levels of performance, followed by Oseltamivir::Influenza, Ethynil-estradiol-/-norgestimate::Acne, and Suprep-Bowel-Prep-Kit::Bowel-Preparation at a second level, and finally Prozac::Depression consistently exhibiting the lowest performance in every classical configuration ($\Delta F1 = .10$ compared to the other use cases). Examining examples of the depression use casem we find a high frequency of opinionated texts where sentiments are clearer but misleading, i.e., people provide either explicit negative or positive expressions, but this is not connected to the target (argumentative, non-argumentative). Moreover, in this use case, due to its psychological nature, perceptions about what are positive or negative effects of the medication are even more subjective than in other use cases were there's only a physical state to be perceived. We also find in this use case that shorter sentences are usually employed, which seem to be more problematic to be correctly classified.

In contrast, transformer-based approaches consistently demonstrated higher performance on average, outperforming classical models in every use case. Here we can observe two levels of performance, with the cases Liraglutide::Diabetes-Type-2, Ethynil-estradiol-/-norgestimate::Acne, Prozac::Depression and Ethinyl-estradiol-/-norgestimate::Birth-Control exhibiting the highest levels of performance, and Suprep-Bowel-Prep-Kit::Bowel-Preparation and Oseltamivir::Influenza exhibiting the lowest performance.

Regarding the ternary classification task, the findings align with those of the binary classification task. In the SGD classifier, once again, the class Liraglutide::Diabetes-Type-2 demonstrates the highest performance, while the class Prozac::Depression exhibits the lowest performance. However, when it comes to transformer-based approaches, the class Liraglutide::Diabetes-Type-2 remains the best performing one, but this time the class Suprep-Bowel-Prep-Kit::Bowel-Preparation displays the lowest performance.

Overall, we find that the Diabetes use case is the easier to classify, while the Depression and Influenza cases are the most difficult ones. ELABORAR POR QUÉ

Task: show plots for different models and performances per use case. - EZEQUIEL: Crees que merece la pena, y si es así, crees que te sería fácil prepararlo? Qué tal la que he añadido? Aporta simplemente para el punto de mencionar qué diferente es la argumentación a nivel de caso de uso y cómo diferentes modelos los capturan de manera diferente. Por ejemplo, depresión es lo más difícil de clasificar para un modelo clásico, sobre todo en 3-classes task. Por otra parte la información que aporta es la misma que la de la tabla, así que no sé, qué opináis?

	2-classes task		3-classes task	
Use case	SGD	BioBERT	SGD	BioBERT
Suprep-Bowel-Prep-Kit::Bowel-Preparation	0.7353	0.7958	0.5672	0.7372
Ethinyl-estradiol-/-norgestimate::Birth-Control	0.7715	0.8243	0.6266	0.7447
Liraglutide::Diabetes,-Type-2	0.8030	0.8557	0.6774	0.7848
Ethinyl-estradiol-/-norgestimate::Acne	0.7417	0.8487	0.5862	0.7542
Prozac::Depression	0.6442	0.8281	0.5147	0.7458
Oseltamivir::Influenza	0.7547	0.7666	0.6196	0.7578

Table 5: Performance of the different approximations per use case (F-1 scores).

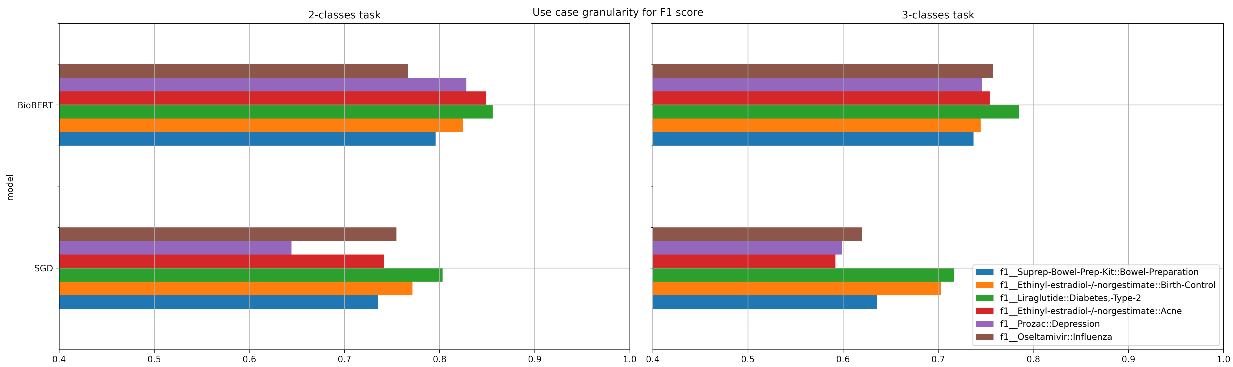


Figure 7: Performance per use case

This suggests that transformer models may be better suited for detecting argumentation structures in text, particularly in cases where the language is more complex and nuanced. Overall, these findings underscore the potential benefits of using transformer models for NLP classification tasks, particularly in the context of argumentation detection.

Eventhough the texts and general domain of the dataset are consistent, when we look closely to different kinds of pharmaceutical products and their cases we find that language, argumentation methods, vocabulary, use of sentiment, text length, and other linguistic features are quite different across use cases. In practice, when we work with different use cases we are almost working with different subdomains. Sentences that don't contain any relevant value in one use case could be extremely relevant in other cases. This is a extra challenge for our classification approaches but also a validity and generalisation check if they succeed.

7. Conclusions

[Translate bullet points into text]

- Classical machine learning models and fine-tuned transformer models provide comparable results for the binary classification with a different of X. Nevertheless, in the three class classification we find that classical
- Binary and ternary classification experiments show as two different tasks: if we compare classification performance for the binary classification problem to the ternary one, we observe that there is no linear tendency between the results of binary and ternary. For instance, some use cases with very high performance in the binary case become low perfomative and viceversa, in comparison. This is the case for "Depression" and "Influenza" in the case of transformers.
- Within the same dataset and general domain, use cases present very different qualitative characteristics: non-relevant examples from one use case could become

- Our definition of use case as a combination of drug and condition seems consistent, in the light that the same drug applied to two different conditions generate very different texts. On the other hand, similar and related conditions – say mental conditions, ginecology products, etc. – constitute consistent subdomains to be considered. In other words, aggregation by condition would make more sense that aggregation by product.
- The qualitatively differences between use cases have served as a challenging validation check to our system, which in the case of transformers has shown high performance in every experiment.
- We have observed noticeable qualitative differences between which examples are correctly classified by classical machine models and which are correctly classified by trasformer models.
- In the binary case, the traditional approach (SGD) offers relatively good results compare to transformer results. The differences across cases are impacting the performance of the classical models that have more difficulty to perform generalisation.
- It is interestingly surprising the good overall performance of simple methods such as
- One of our task design decision was to simplify the problem of detecting relations between the argumentative elements towards the *major claim* and turn it into a ternary classification problem. This has been challenging in the past with traditional methods – and we can see the results that we obtained for that schema ourselves – but with modern techniques this has become a feasible and simple approximation to the problem.
- This "simplification" of the argumentation scheme is not applicable to every argumentation problem. In those domains with higher complexity of argumentation where we find multiple *major claims* and multiple argumentation unit types and relations, this approach might find difficulties. Nevertheless,
- There are challenges intrisic to the definition of the argumentation scheme ... Difficulties distinguishing when are we talking of one product or a previous product, distinghuish anecdote from current experience and
- Redefinitions of the argumentaton scheme can allow us to approach this problems as pure classification problems if we find the right argumentation schema.
- The role of context of individual sentences within one review and simultaneously reviews within use cases seems a critical source of information to address this problems. In this work we have centered our efforts on classification systems that use annotation schemes . **Structures and sequences**
- **Linguistic features** that seem to have little influence
- **Challenges of the dataset** One of the challenges of the dataset is the flexibility of what an argument is, depending on the use case.
- **Anecdotal evidence** is a key aspect of this dataset. Argumentation could take many shapes content-wise
- Further work is necessary to understand
- In the ternary case,
- The gap bet

Acknowledgments

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